

Genetic report for longevity

Peripheral Artery Disease

Polygenic risk of leg disease, 9p21, Lp(a), tobacco (CHRNA3/5), thrombosis (F5), diabetes (TCF7L2), lipids and atherosclerotic context

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Peripheral arterial disease (PAD) is atherosclerosis of the arteries of the legs. Sharing genetics with coronary disease, but the importance of factors changes: tobacco and diabetes count much more in PAD than in the heart, and Lp(a) and thrombosis also count. The estimated heritability goes through 20–30%. This report reads three PAD polygenic scores, a GWAS loci score (the GWAS of the Million Veteran Program and previous meta-analyses), the main loci (9p21, LPA, CHRNA3/5, SH2B3/ATXN2, HDAC9, COL4A1, EDNRA, F5, TCF7L2) and joins the risk factors scores already calculated in other reports (coronary disease, LDL, ApoB, Lp(a), diabetes, tobacco). A genetic score is not diagnostic; PAD is rare before the age of 50 and depends mainly on modifiable factors.

Only genes: genotyping chip + local imputation + polygenic scores (PGS Catalog) and loci scores of GWAS Catalog, with percentiles in the reference population. Each theme is divided into favourable and unfavourable and linked to the aging mechanisms of the list of objectives. It is neither a diagnosis nor a prescription. All practical ideas are marked as a suggestion (not applied).

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Summary

18

variants

14

in the chip

4

imputed
($DR2 \geq 0.8$)

0

not found

4

Scores with
percentile

Overall reading. The signs are divided, but the most complete scores point to a low to typical risk. The multi-trait score of 2025 (970 thousand variants, the most predictive) is at the 22nd percentile and the 29 loci score of GWAS at the 0th percentile (below all 633 reference people; few risk alleles outside the 9p21). On the contrary, the two Biobank UK scores for "peripheral vascular disease" (unspecific diagnostic code) are at the 81 and 89 percentiles. The clear unfavourable point is 9p21 in risk homozygosity (rs1333049 C/C and rs1537372 T/T): it is the strongest common locus for coronary disease and PAD and acts on the arterial wall, not on lipids. In favor: LPA without high Lp(a) alleles (rs10455872 A/A, rs3798220 T/T), without factor V Leiden, SH2B3 C/C (without inflammatory allele), ATXN2, COL4A1, NFAT5, PTPN11 and CELSR2 on the favourable side. CHRNA3 and CHRNA5 with a copy (only tobacco), TCF7L2 C/T and IL6R C/A (a typical protective copy). The context of the other reports is favourable: coronary disease (metaGRS) at the 5th percentile, LDL in 20, HbA1c in 7 and PGS of having smoked at 1.6; Polygenic Lp(a) in 75 (in the limit) and triglycerides in 68. The two major factors of PAD, tobacco and diabetes, are on the favourable side.

Favourable

- **PAD (multi-trait, 969 771 variants):** the 22nd percentile
- **PAD (loci of GWAS):** the 0th percentile
- **Coronary disease (metaGRS, PGS000018):** the 5th percentile (Advanced Cardiovascular Report)
- **LDL (PGS000892):** the 20th percentile (Lipid Report in Detail)
- **HbA1c (PGS001953):** the 7th percentile (Glycosis, Insulin and Diabetes reports)
- **Have smoked (974 variants):** the 2nd percentile (Caffeine Report and Stimulants)
- **LPA rs10455872 AA:** LPA: the G allele marks small isoforms and high Lp(a); it is from the strongest loci to PAD (effect greater than in coronary disease).
- **LPA rs3798220 TT:** LPA I4399M: C allele is associated with high Lp(a) and more arterial disease.
- **SH2B3 rs3184504 CC:** SH2B3 R262W: T allele (Trp) increases inflammation and platelets; risk of coronary disease and PAD.
- **F5 (Leiden) rs6025 CC:** Factor V Leiden: the T allele gives resistance to activated protein C; risk of thrombosis and, more modestly, PAD and occlusion of arterial bridges.
- **ATXN2 rs4766578 AA:** ATXN2/SH2B3 (12q24): pleiotropic locus of inflammation, platelets and arterial disease.
- **COL4A1 rs1975514 TT:** COL4A1: collagen IV of the basal membrane of the vessels; arterial disease and small cerebral vessels.
- **NFAT5 rs244415 AA:** NFAT5: transcription factor of osmotic and inflammatory response of the vascular wall.
- **PTPN11 rs11066301 AA:** PTPN11 (12q24): growth factor signaling; neighbor of SH2B3.
- **CELSR2 rs7528419 GA:** CELSR2/SORT1 (1p13): the large locus of LDL; enters the cholesterol pathway.

Unfavourable

- **Peripheral vascular disease (LDpred2, 599514 variants):** the 81st percentile
- **Peripheral vascular disease (242 variants):** the 89th percentile
- **Lp(a) (PGS000752):** the 75th percentile (Lipid Report in Detail)
- **CDKN2B-AS1 (9p21) rs1333049 CC:** 9p21: the C allele is at risk for coronary disease, PAD and aneurysm; each copy sum ~15–30% of relative risk
- **CDKN2B-AS1 (9p21) rs1537372 TT:** 9p21: main variant in GWAS of PAD (Million Veteran Program); marks the same block of rs1333049.

Here, the scores out of the typical range (≤ 25 or ≥ 75 , in the trait direction) and the variants with strong or moderate evidence. The weak evidence variants are in the tables of each theme.

Strand: All genotypes are in the strand + (forward) of GRCh38, the same as dbSNP/Ensembl. Of the 14 chip genotypes, 0 needed complement, 1 are palindromic (A/T or C/G; orientation confirmed by the position and reference alleles of the chip, which is already in the strand +) and 0 were read by the position (merged rsID). The imputations come from the local VCF, also in the strand +. The cited effect alleles of studies that use the strand of the gene were converted to the strand +; therefore, some genotypes appear with different letters from those in the literature (e.g., genes in the strand – such as COL1A1 or VDR).

Polygenic risk of peripheral arterial disease

Three PAD scores from PGS Catalog (242 variants and 600 thousand variants from the same study of UK Biobank; 970 thousand variants of a multi-trait score of 2025) and a locus score of GWAS Catalog. Multi-trait scores gather information on coronary disease, lipids, diabetes and tobacco and are the best predictors.

Ageing mechanisms: Vascular Aging; Arterial stiffness; Chronic Inflammation

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
PAD (multi-trait, 969 771 variants)	Favourable	the 22nd percentile · z=-0.74	Higher score : higher risk. Multi-trait 2025 score, the most predictive of the three. PGS012546 (PGS Catalog; Multi-trait PRS; Peripheral arterial disease; PGP000790, PMID 41800596). Variants with genotype: 949764 of 969771 (covering of 98.5% coefficients).
Peripheral vascular disease (LDpred2, 599514 variants)	Unfavourable	the 81st percentile · z=0.91	Higher score : higher risk. PGS002055 (PGS Catalog; LDpred2 (bigsnpr); Peripheral vascular disease, unspecified; PGP000263, PMID 34995502). Variants with genotype: 587482 of 599514 (covering of 97.0% coefficients).
Peripheral vascular disease (242 variants)	Unfavourable	the 89th percentile · z=1.21	Higher score : higher risk. Few variants and low power. PGS001843 (PGS Catalog; Penalized regression (bigstatsr); Peripheral vascular disease, unspecified; PGP000263, PMID 34995502). Variants with genotype: 232 of 242 (covering of 95.5% coefficients).
PAD (loci of GWAS)	Favourable	the 0th percentile · z=-3.20	Higher score : higher risk. Loci score of GWAS Catalog (p≤510-8; 1 variant by 500 kb window; simple count of risk alleles, without weighting, because the studies use different scales: 29 of 29 loci with genotype. Studies: Guindo-Martinez M (Peripheral vascular disease); Klarin D (Peripheral artery disease); Matsukura M (Peripheral artery disease); Sakaue S (Peripheral artery disease); Verma A (Peripheral vascular disease); Verma A (Peripheral vascular disease, unspecified (PheCode 443.9)); van Zuydam N (Peripheral artery disease). Variants with genotype: 29 of 29 (covering of 100.0% coefficients).

Favourable

- PAD (multi-trait, 969 771 variants) (the 22nd percentile)
- PAD (GWAS loci) (the 0th percentile)

Unfavourable

- Peripheral vascular disease (LDpred2, 599 514 variants) (the 81st percentile)
- Peripheral vascular disease (242 variants) (the 89th percentile)

Main loci of peripheral atherosclerosis

9p21 (CDKN2B-AS1, the strongest locus for coronary disease and PAD), LPA (lipoprotein(a)), SH2B3/ATXN2 (inflammation and blood cells), HDAC9 (arterial calcification), COL4A1 (basal membrane of vessels), EDNRA (endothelin receptor), NFAT5, PTPN11, CELSR2/SORT1 (LDL) and ABO.

Ageing mechanisms: Vascular Aging; Arterial stiffness; Remodelling of the extracellular matrix

Favourable

- LPA rs10455872 AA
- LPA rs3798220 TT
- SH2B3 rs3184504 CC
- ATXN2 rs4766578 AA
- COL4A1 rs1975514 TT
- NFAT5 rs244415 AA
- PTPN11 rs11066301 AA
- CELSR2 rs7528419 GA

Unfavourable

- CDKN2B-AS1 (9p21) rs1333049 CC
- CDKN2B-AS1 (9p21) rs1537372 TT

Favourable variants (8)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
LPA	rs10455872 6:160589086 · A/G	AA	chip	G (0; 7%)	strong	LPA: the G allele marks small isoforms and high Lp(a); it is from the strongest loci to PAD (effect greater than in coronary disease).
LPA	rs3798220 6:160540105 · T/C	TT	chip	C (0; 1%)	strong	LPA I4399M: C allele is associated with high Lp(a) and more arterial disease.
SH2B3	rs3184504 12:111446804 · T/A/C/G	CC	chip	T (0; 46%)	moderate	SH2B3 R262W: T allele (Trp) increases inflammation and platelets; risk of coronary disease and PAD.
ATXN2	rs4766578 12:111466567 · T/A	AA	chip	T (0; 48%)	moderate	ATXN2/SH2B3 (12q24): pleiotropic locus of inflammation, platelets and arterial disease. Meaning: GWAS Catalog: Peripheral vascular disease (Verma A 2024; p=5e-24); T ↑ allele
COL4A1	rs1975514 13:110176544 · T/C/G	TT	imputation DR2=0.99	C (0; 36%)	moderate	COL4A1: collagen IV of the basal membrane of the vessels; arterial disease and small cerebral vessels. Meaning: GWAS Catalog: Peripheral artery disease (van Zuydam N 2021; p=2e-23); C ↑ allele
NFAT5	rs244415 16:69632780 · G/A	AA	imputation DR2=0.99	G (0; 58%)	moderate	NFAT5: transcription factor of osmotic and inflammatory response of the vascular wall. Meaning: GWAS Catalog: Peripheral vascular disease (Verma A 2024; p=3e-15); G ↑ allele
PTPN11	rs11066301 12:112433568 · A/G	AA	chip	G (0; 43%)	moderate	PTPN11 (12q24): growth factor signaling; neighbor of SH2B3. Meaning: GWAS Catalog: Peripheral artery disease (van Zuydam N 2021; p=3e-12); G ↑ allele
CELSR2	rs7528419 1:109274570 · A/G	GA	chip	A (1; 79%)	moderate	CELSR2/SORT1 (1p13): the large locus of LDL; enters the cholesterol pathway. Meaning: GWAS Catalog: Peripheral artery disease (van Zuydam N 2021; p=2e-12); allele A ↑

Unfavourable variants (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying;	Evidence	Interpretation
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frequency in reference)						
CDKN2B-AS1 (9p21)	rs1333049 9:22125504 · G/A/C	CC	chip	C (2; 47%)	strong	9p21: the C allele is at risk for coronary disease, PAD and aneurysm; each copy adds ~15–30% of relative risk. The mechanism is smooth muscle proliferation and inflammation of the arterial wall, not lipids.
CDKN2B-AS1 (9p21)	rs1537372 9:22103184 · G/A/T	TT	chip	T (2; 44%)	strong	9p21: main variant in GWAS of PAD (Million Veteran Program); marks the same block of rs1333049. Meaning: GWAS Catalog: Peripheral artery disease (van Zuydam N 2021; p=7e-41); T ↑ allele

Typical / intermediate variants (4)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
CHRNA3	rs10851907 15:78623522 · G/A/C	GA	chip	A (1; 43%)	moderate	CHRNA3: cluster of nicotinic receptors; the locus of PAD more linked to tobacco. Meaning: GWAS Catalog: Peripheral artery disease (van Zuydam N 2021; p=1e-35); allele A ↑
EDNRA	rs72957606 4:147478859 · A/G	AA	imputation DR2=1.00	G (0; 15%)	moderate	EDNRA: endothelin-1 receptor (vasoconstriction, smooth muscle proliferation). Meaning: GWAS Catalog: Peripheral vascular disease, unspecified (PheCode 443.9) (Verma A 2024; p=3e-18); allele A ↓
HDAC9	rs2107595 7:19009765 · G/A	GG	chip	A (0; 17%)	moderate	HDAC9-TWIST1: vascular calcification; Stroke of large arteries and PAD. Meaning: GWAS Catalog: Peripheral vascular disease, unspecified (PheCode 443.9) (Verma A 2024; p=1e-15); G ↓ allele
ABO	rs115478735 9:133274295 · A/T	AA	imputation DR2=1.00	T (0; 18%)	moderate	ABO: non-O blood group increases the factor VIII and von Willebrand factor; thrombosis and arterial disease. Meaning: GWAS Catalog: Peripheral vascular disease (Verma A 2024; p=8e-12); allele A ↓

Tobacco, diabetes and thrombosis

The specific loci of PAD (compared to coronary disease) are tobacco (CHRNA3/CHRNA5) and coagulation (F5 Leiden); TCF7L2 comes from diabetes. Without tobacco, the effect of CHRNA3/5 practically does not apply. IL6R Asp358Ala (C allele) reduces IL-6 signaling and is associated with less arterial disease.

Ageing mechanisms: Vascular Aging; Insulin resistance; Chronic Inflammation

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Have smoked (974 variants)	Favourable	the 2nd percentile	Tobacco is the biggest risk factor of PAD; very low predisposition and never smoked. Source: Caffeine report and Stimulants.

Favourable

- Have smoked (974 variants) (the 2nd percentile)
- F5 (Leiden) rs6025 CC

Unfavourable

None.

Favourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
F5 (Leiden)	rs6025 1:169549811 · C/A/T	CC	chip	T (0; 1%)	strong	Factor V Leiden: the T allele gives resistance to activated protein C; risk of thrombosis and, more modestly, PAD and occlusion of arterial bridges.

Typical / intermediate variants (3)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
CHRNA5	rs16969968 15:78590583 · G/A	GA	chip	A (1; 37%)	strong	CHRNA5 D398N: allele A increases tobacco consumption and dependence; the effect on PAD passes almost all through tobacco.
TCF7L2	rs7903146 10:112998590 · C/G/T	CT	chip	T (1; 32%)	strong	TCF7L2: T allele is the largest common locus of diabetes type 2; it enters PAD through diabetes.
IL6R	rs2228145 1:154454494 · A/C/T	CA	chip	A (1; 64%)	moderate	IL6R Asp358Ala: C allele (Ala) increases the soluble receptor and reduces IL-6 signaling in cells; it is associated with less coronary disease and PAD. The unfavourable allele is A.

Genetic risk factors (from other reports)

Scores already calculated in the Advanced Cardiovascular Reports, Lipids in Detail, Glucose, Insulin and Diabetes and Caffeine and Stimulants, gathered here as a context. They were not recalculated.

Ageing mechanisms: Vascular Aging; Arterial stiffness; Insulin resistance

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Coronary disease (metaGRS, PGS000018)	Favourable	the 5th percentile	Shares most of the genetics with PAD. Source: Advanced Cardiovascular Report.
Coronary disease (GPS_Mult, PGS003725)	Typical / intermediate	the 30th percentile	"typical" label. Source: Advanced Cardiovascular Report.
LDL (PGS000892)	Favourable	the 20th percentile	Lower LDL is better; percentile 20.5 (typical tag in the origin report, at the limit of favourable). Source: Lipid Report in Detail.
ApoB (PGS002102)	Typical / intermediate	the 46th percentile	"typical" label. Source: Lipid Report in Detail.
Lp(a) (PGS000752)	Unfavourable	the 75th percentile	It labels "typical" in the original report (in the limit); Lp(a) counts more in PAD than in coronary disease. Source: Lipid Report in Detail.
2 type diabetes (PGS000014)	Typical / intermediate	the 57th percentile	"typical" label. Source: Glucose, Insulin and Diabetes report.
HbA1c (PGS001953)	Favourable	the 7th percentile	"Favourable" label. Source: Glucose, Insulin and Diabetes report.
Triglycerides (PGS002719)	Typical / intermediate	the 68th percentile	"typical" label. Source: Lipid Report in Detail.

Favourable

- Coronary disease (metaGRS, PGS000018) (the 5th percentile)
- LDL (PGS000892) (the 20th percentile)
- HbA1c (PGS001953) (the 7th percentile)

Unfavourable

- Lp(a) (PGS000752) (the 75th percentile)

Mechanisms of ageing

Linking the results to the aging mechanisms of the list of objectives for immortality. Sorted by the number of unfavourable results.

Mechanism	Unfavourable	Favourable	Typical
Vascular Aging	Peripheral vascular disease (LDpred2, 599 514 variants) (the 81st percentile); Peripheral vascular disease (242 variants) (the 89th percentile); CDKN2B-AS1 (9p21) CC; CDKN2B-AS1 (9p21) TT; Lp(a) (PGS000752) (the 75th percentile)	PAD (multi-trait, 969 771 variants) (the 22nd percentile); PAD (GWAS loci) (the 0th percentile); LPA AA; LPA TT; SH2B3 CC; ATXN2 AA; COL4A1 TT; NFAT5 AA; PTPN11 AA; CELSR2 GA; Have smoked (974 variants) (the 2nd percentile); F5 (Leiden) CC; Coronary disease (metaGRS, PGS000018) (the 5th percentile); LDL (PGS000892) (the 20th percentile); HbA1c (PGS001953) (the 7th percentile)	CHRNA3 GA; EDNRA AA; HDAC9 GG; ABO AA; CHRNA5 GA; TCF7L2 CT; IL6R CA; Coronary disease (GPS_Mult, PGS003725) (the 30th percentile)
Arterial stiffness	Peripheral vascular disease (LDpred2, 599 514 variants) (the 81st percentile); Peripheral vascular disease (242 variants) (the 89th percentile); CDKN2B-AS1 (9p21) CC; CDKN2B-AS1 (9p21) TT; Lp(a) (PGS000752) (the 75th percentile)	PAD (multi-trait, 969 771 variants) (the 22nd percentile); PAD (GWAS loci) (the 0th percentile); LPA AA; LPA TT; SH2B3 CC; ATXN2 AA; COL4A1 TT; NFAT5 AA; PTPN11 AA; CELSR2 GA; Coronary disease (metaGRS, PGS000018) (the 5th percentile); LDL (PGS000892) (the 20th percentile); HbA1c (PGS001953) (the 7th percentile)	CHRNA3 GA; EDNRA AA; HDAC9 GG; ABO AA; Coronary disease (GPS_Mult, PGS003725) (the 30th percentile); ApoB (PGS002102) (the 46th percentile); 2 (PGS000014) type diabetes (the 57th percentile); Triglycerides (PGS002719) (the 68th percentile)
Chronic Inflammation	Peripheral vascular disease (LDpred2, 599 514 variants) (the 81st percentile); Peripheral vascular disease (242 variants) (the 89th percentile)	PAD (multi-trait, 969 771 variants) (the 22nd percentile); PAD (GWAS loci) (the 0th percentile); Have smoked (974 variants) (the 2nd percentile); F5 (Leiden) CC	CHRNA5 GA; TCF7L2 CT; IL6R CA
Remodelling of the extracellular matrix	CDKN2B-AS1 (9p21) CC; CDKN2B-AS1 (9p21) TT	LPA AA; LPA TT; SH2B3 CC; ATXN2 AA; COL4A1 TT; NFAT5 AA; PTPN11 AA; CELSR2 GA	CHRNA3 GA; EDNRA AA; HDAC9 GG; ABO AA
Insulin resistance	Lp(a) (PGS000752) (the 75th percentile)	Have smoked (974 variants) (the 2nd percentile); F5 (Leiden) CC; Coronary disease (metaGRS, PGS000018) (the 5th percentile); LDL (PGS000892) (the 20th percentile); HbA1c (PGS001953) (the 7th percentile)	CHRNA5 GA; TCF7L2 CT; IL6R CA; Coronary disease (GPS_Mult, PGS003725) (the 30th percentile); ApoB (PGS002102) (the 46th percentile); 2 (PGS000014) type diabetes (the 57th percentile); Triglycerides (PGS002719) (the 68th percentile)

Practical part

Ideas to protect the arteries of the legs, crossed with the genetic plan v2 and with the current medication (retatrutide, dapagliflozin, telmisartan, tadalafil, rapamycin, lithium 5 mg, low dose naltrexone, isotretinoin). Everything is suggestion (not applied). With 9p21 of risk in homozygosity and the rest favourable, what counts most is to keep the modifiable factors where they are.

1. Continue without tobacco, vaping or smoking

Genetics: CHRNA3/CHRNA5 with 1 copy; PGS of having smoked p1.6; SERPINA1 PiMZ (Lung Report)

Tobacco multiplies the risk of PAD several times (more than in coronary disease) and adds to PiMZ in the lung. Avoid passive smoking and electronic cigarettes.

V2 genetic plan and medication: Compatible with the plan. • *suggestion (not applied)*

2. Walking and regular aerobic training

Genetics: 9p21 C/C (arterial wall); HDAC9 and COL4A1

Walking regularly (including ascents or intervals) is the most evident intervention for the circulation of the legs and improves the endothelium function. It joins the strength training of the plan.

V2 genetic plan and medication: It is compatible with the plan (aerobic exercise and force). • *suggestion (not applied)*

3. LDL and ApoB low

Genetics: CELSR2/SORT1 favourable; LDL p20; Lp(a) polygenic p75

In PAD, as in coronary disease, the lower the LDL and ApoB, the better. Maintain high fiber and moderate saturated fat; follow LDL and ApoB in routine analysis.

V2 genetic plan and medication: Compatible with the plan (LDL/ApoB: the lower, the better; the more fiber). • *suggestion (not applied)*

4. Omega-3 at the dose of the plan

Genetics: 9p21 C/C; p68 triglycerides

The fish oil that already takes, at the dose of the plan (3.2–4 g/day of EPA+DHA), low triglycerides; the trial with high dose EPA reduced arterial events.

V2 genetic plan and medication: Already underway (fish oil); dose of the plan 3.2–4 g/day. • *suggestion (not applied)*

5. Know the alarm signal

Genetics: Multi-trait p22 score, but risk 9p21

Pain or cramp in the belly of the leg (or thigh/buttock) when walking, which passes at rest, is the typical symptom of PAD. If it appears, talk to the doctor.

V2 genetic plan and medication: No change in the plan. • *suggestion (not applied)*

Avoiding or with caution

- Smoking in any form, electronic cigarettes and passive smoking. *suggestion (not applied)*
- Long periods without physical activity. *suggestion (not applied)*
- Oversaturated fat (sobe LDL and ApoB). *suggestion (not applied)*

Limitations

- PAD scores were developed mainly in older people and with many smokers and diabetics; in those who never smoked and have low genetic glycemia, the predictive value is lower.
- The multi-trait score includes information on coronary disease, lipids and tobacco; it is not independent of context scores.
- The Lp(a) depends mainly on the number of repetitions KIV-2 of LPA, which is not read by chip; SNPs and scores capture only part.
- The percentiles compare with a reference population of 633 genomes; the effects and percentiles may not be transferred between populations.
- Only genes: does not include analyzes, tests or lifestyle factors. It is not a diagnosis.

Methods and sources

Genotypes: first the genotyping chip (GRCh38, strand +), by rsID and, for cast rsIDs, by the position; then the local imputation (Beagle 5.5 with a public reference panel of complete genomes), only with $DR2 \geq 0.8$. The variants that do not pass are "SNP not found in the data". The raw DNA was processed locally; to the public databases only rsIDs were sent.

Variation classification: The number of copies of the unfavourable allele is counted and compared to what was expected in the reference population: at least 0.5 copies above that expected = unfavourable; at least 0.5 below = favourable; the rest = typical/intermediate. For rare variants of large effect, having 0 copies = favourable and having ≥ 1 = unfavourable. The unfavourable alleles come from the literature (cured) or, when indicated, from the strongest association in GWAS Catalog.

Main variants per gene: for genes labeled as "GWAS Catalog", the variant with the strongest genomic association ($p \leq 5 \times 10^{-8}$) is used, mapped to this gene and with available genotype.

Polygenic scores (PGS Catalog): Harmonized coefficients for GRCh38 applied to the imputed genome (chip + $DR2 \geq 0.8$, dosage). Empirical percentile in the reference population ($n=633$), calculated with the same variants. List: PGS012546 (PGS Catalog; Multi-trait PRS; Peripheral arterial disease; PGP000790, PMID 41800596 · PGS002055 (PGS Catalog; LDpred2 (bigsnpr); Peripheral vascular disease, unspecified; PGP000263, PMID 34995502 · PGS001843 (PGS Catalog; Penalized regression (bigstatsr); Peripheral vascular disease, unspecified; PGP000263, PMID 34995502).

GWAS Catalog loci scores: for traits without PGS, weighted sum of risk variants ($p \leq 5 \times 10^{-8}$, one per window of 500kb), with a percentile calculated in the same way.

Percentiles: ≤ 25 or ≥ 75 (in the direction of the trait) = favourable or unfavourable; between 25 and 75 = typical.

Main sources

- Verma A et al. Diversity and scale: genetic architecture of 2068 traits in the VA Million Veteran Program (Science 2024).
- van Zuydam N et al. GWAS of peripheral arterial disease (Circ Genom Precision Med 2021).
- Klarin D et al. Genome-wide association study of peripheral artery disease in the Million Veteran Program (Nat Med 2019; PMID 31285632).
- PGS Catalog: PGS001843 and PGS002055 (Privé F et al., Am J Hum Genet 2022; PMID 34995502); PGS012546 (PMID 41800596).
- Helgadottir A et al. 9p21 and arterial disease (Nat Genet 2008).
- Clarke R et al. Variants of LPA and coronary artery disease (N Engl J Med 2009).
- IL6R Genetics Consortium. The interleukin-6 receptor as a target for prevention of coronary heart disease (Lancet 2012).
- GWAS Catalog (NHGRI-EBI), full download of associations.

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